

TP4

Networks and trees

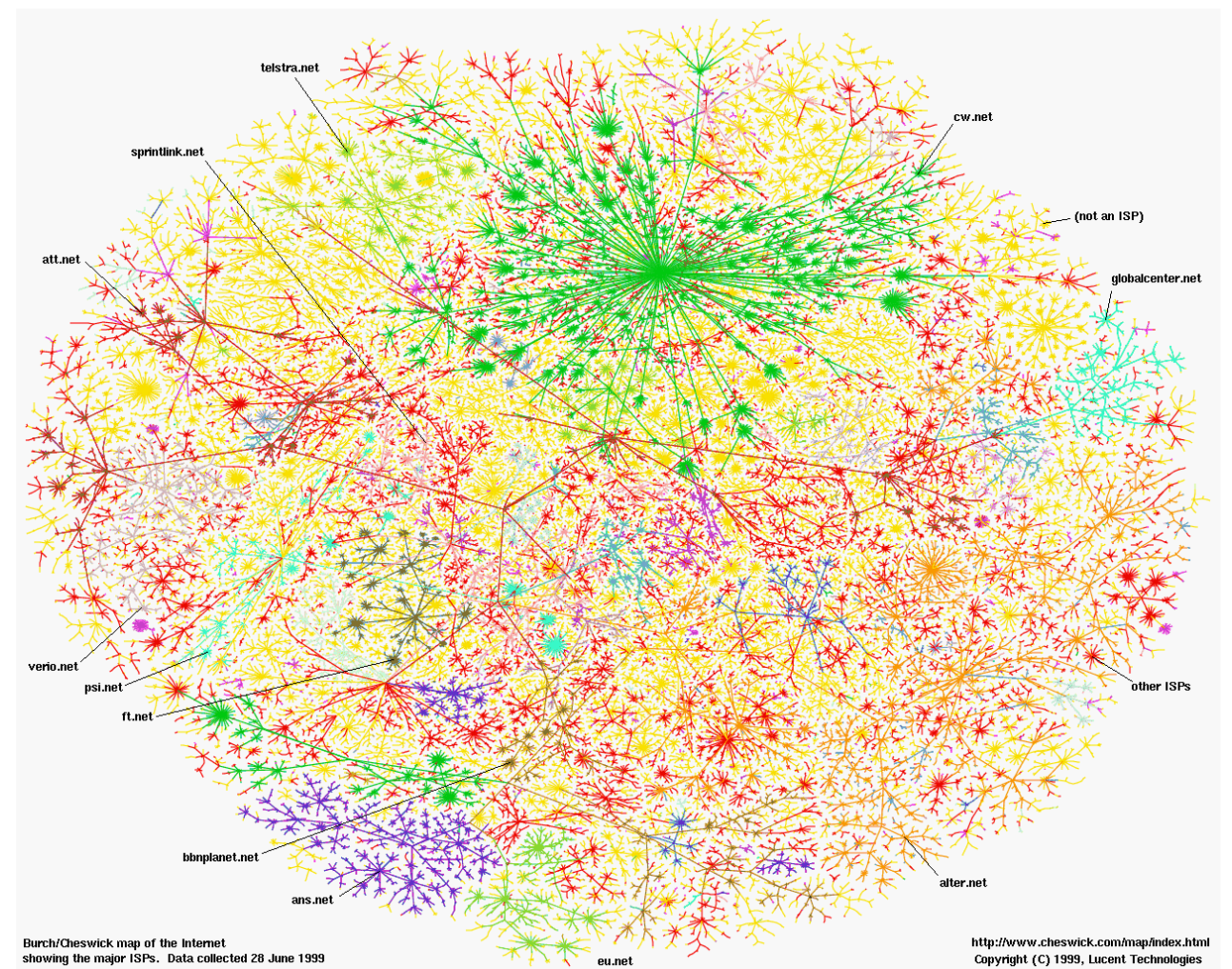
MECD 2024/25

Advanced Data Visualization

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Networks

- A diagram or a map that shows how entities are interconnected;
- Use node-link representation, where nodes represent entities and links (edges) represent their relationship;
- Layout is important (e.g., radial, force-directed, sequential, etc.);
- In plotly we will use `networkx` library, so please `pip install networkx`



Map of the Internet, Bill Cheswick and Hal Burch, 1999

Network

Common code for all the exercises/examples

```
import networkx as nx
```

```
G = nx.random_geometric_graph(200, 0.125)  
pos = nx.spring_layout(G)
```

```
edge_x = []  
edge_y = []  
for edge in G.edges():  
    x0 = pos[edge[0]][0]  
    y0 = pos[edge[0]][1]  
    x1 = pos[edge[1]][0]  
    y1 = pos[edge[1]][1]  
    edge_x.append(x0)  
    edge_x.append(x1)  
    edge_x.append(None)  
    edge_y.append(y0)  
    edge_y.append(y1)  
    edge_y.append(None)
```

```
node_x = []  
node_y = []  
for node in G.nodes():  
    x = pos[node][0]  
    y = pos[node][1]  
    node_x.append(x)  
    node_y.append(y)
```

- Create a graph with 200 nodes and spring layout applied;
- Extract the x and y coordinates for both ends of each edge;
- Extract the x and y coordinate for each node;
- Store positional data in arrays;
- Many ways of reproducing the same result. Please feel free to explore networkx documentation: <https://networkx.org/documentation/stable/reference/index.html>

Network

Plot the graph using networkx data

```
import plotly.graph_objects as go

edge_trace = go.Scatter(
    x=edge_x, y=edge_y,
    line=dict(width=0.5, color='#888'),
    mode='lines',
    name="Edges"
)

node_trace = go.Scatter(
    x=node_x, y=node_y,
    mode='markers',
    hoverinfo='text',
    marker=dict(color='black'),
    name="Nodes"
)

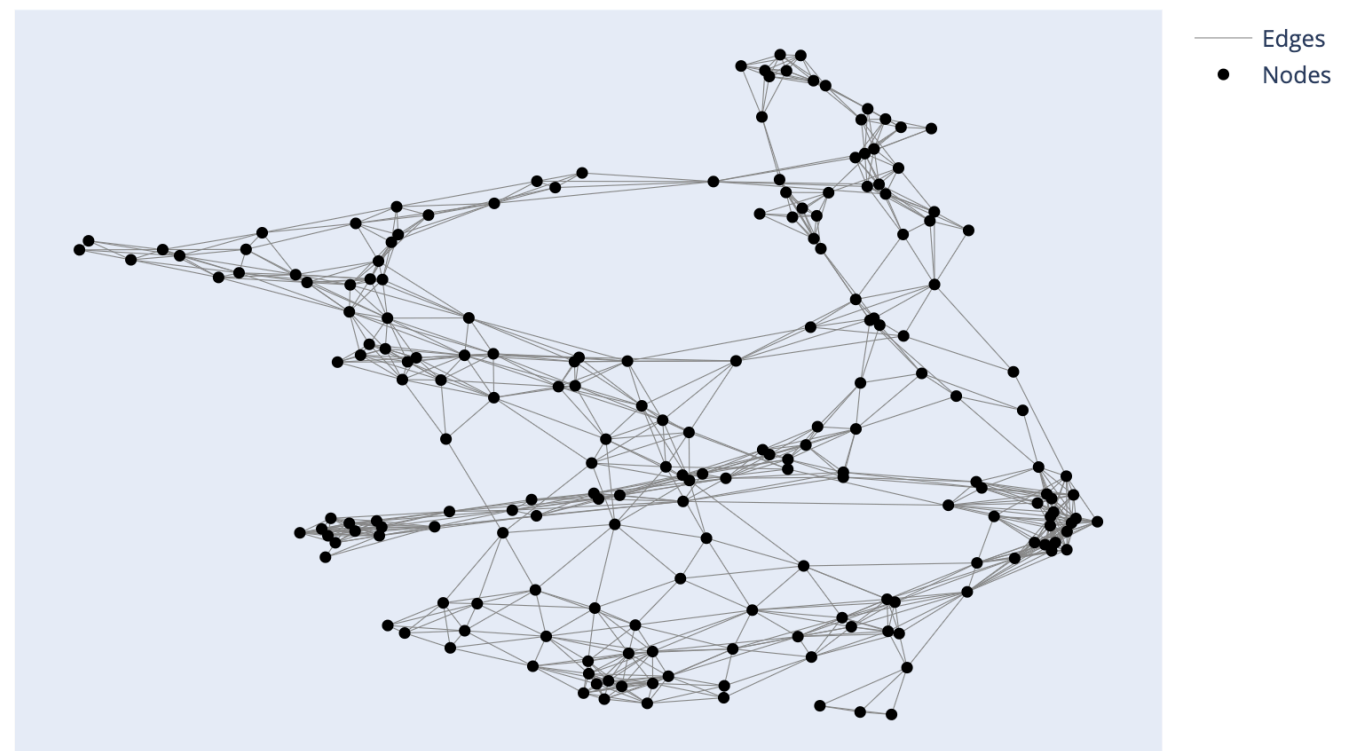
layout = layout=go.Layout(
    title='Random network graph',
    margin=dict(b=20, l=5, r=5, t=40),
    xaxis=dict(showgrid=False,
                zeroline=False,
                showticklabels=False),
    yaxis=dict(showgrid=False,
                zeroline=False,
                showticklabels=False)
)

fig = go.Figure(data=[edge_trace,
                    node_trace], layout=layout)

fig.show()
```

- Use two scatter plot traces—edges and nodes;
- xaxis and yaxis:
 - **showgrid** and **showticklabels** True or False
 - **zeroline**: draw or not the zero line of the axis.

Random network graph



Network

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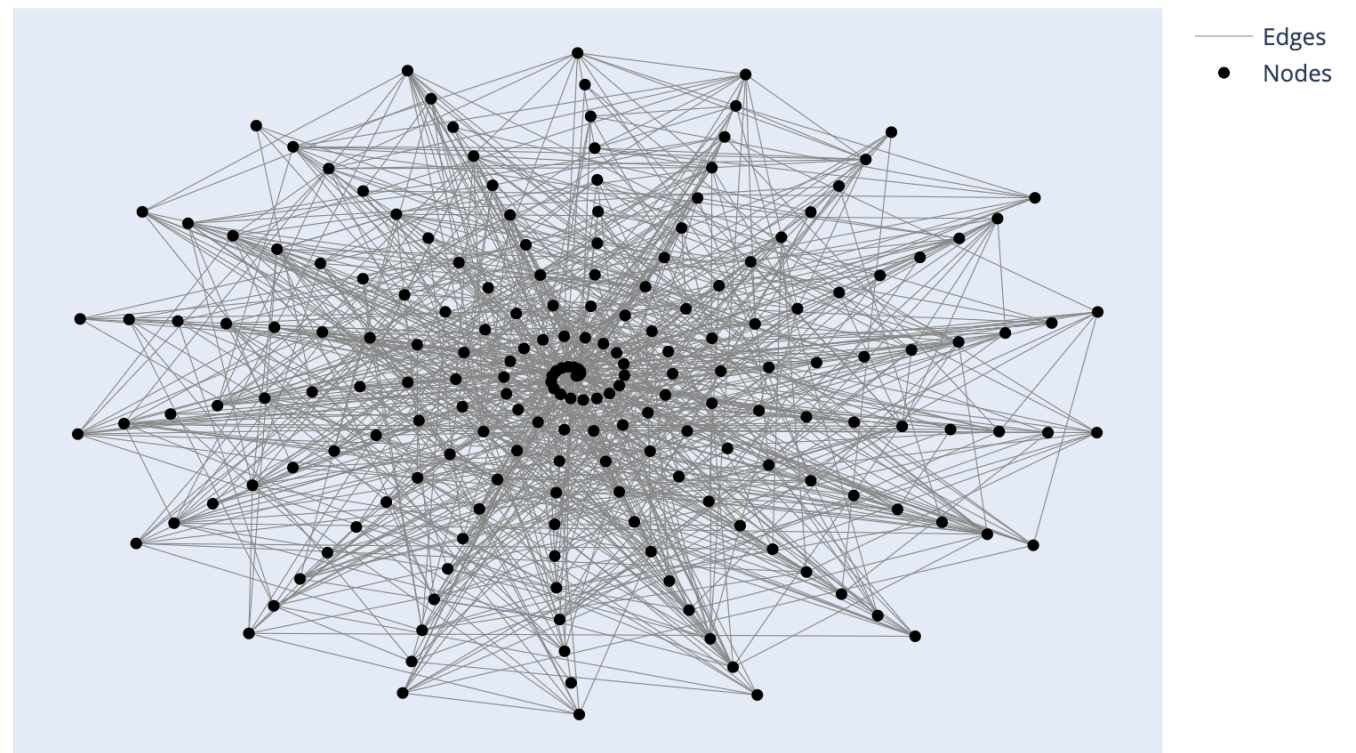
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Random network graph



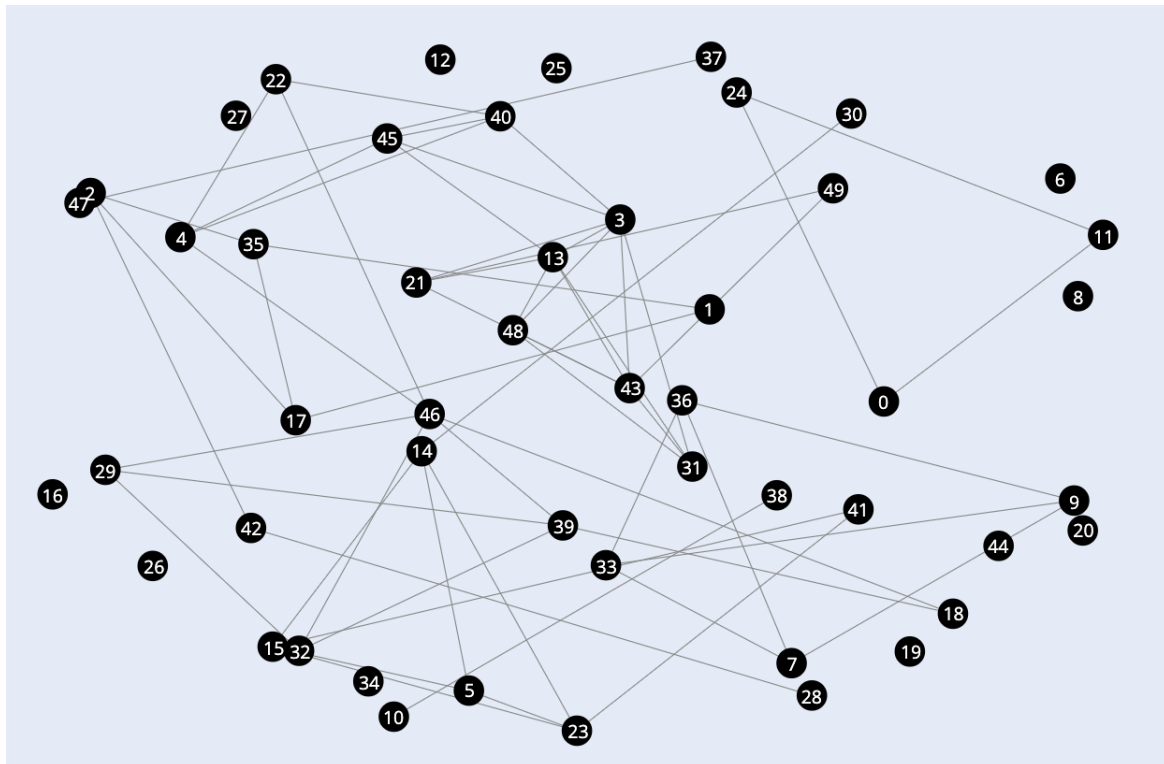
Network

Annotations

```
annotations = []
for k in range(len(node_x)):
    annotations.append(
        dict(
            text = k,
            x = node_x[k], y = node_y[k],
            font_color='rgb(255, 255, 255)',
            font_size=9,
            showarrow=False
        )
    )
```

```
layout.annotations = annotations
```

- `layout.annotations` is used to place a text element anyway in the plot
 - `font` is used to specify font related attributes, such as, size, family, color, etc.
 - **`showarrow`** indicates whether to place arrow or not
- Annotations will be covered in more detail later in this semester.



Questions?

References

- networkx documentation: <https://networkx.org/documentation/stable/reference/index.html>
- Layout annotations reference: <https://plotly.com/python/reference/layout/annotations/>
- More Graph examples: <https://plotly.com/python/network-graphs/>
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